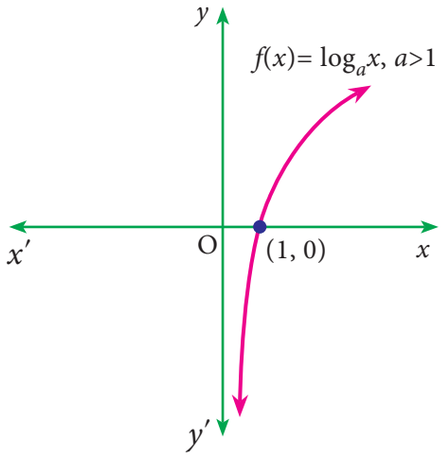


Fig 5.8

Case (ii) when $0 < a < 1$ and $x > 0$

$$f(x) = \log_a x = \begin{cases} > 0 & \text{if } 0 < x < 1 \\ = 0 & \text{if } x = 1 \\ < 0 & \text{if } x > 1 \end{cases}$$

We noticed that x increases, the value of $f(x)$ decreases.

$\therefore$ The graph is as shown in the figure.

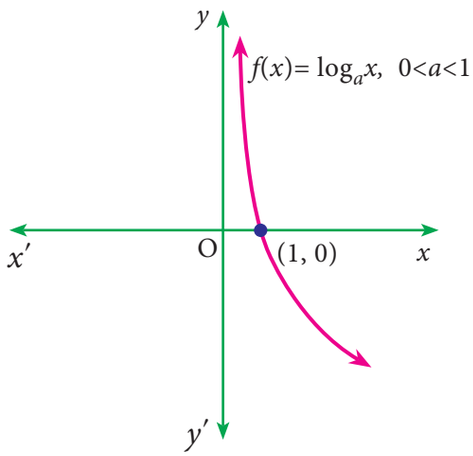


Fig 5.9

Exercise 5.1

1. Determine whether the following functions are odd or even?

   (i) $f(x) = \left(\dfrac{a^x - 1}{a^x + 1}\right)$

   (ii) $f(x) = \log\left(x^2 + \sqrt{x^2 + 1}\right)$

   (iii) $f(x) = \sin x + \cos x$

   (iv) $f(x) = x^2 - |x|$

   (v) $f(x) = x + x^2$

2. Let f be defined by $f(x) = x^3 - kx^2 + 2x$, $x \in R$. Find k, if 'f' is an odd function.

3. If $f(x) = x^3 - \dfrac{1}{x^3}$, $x \neq 0$, then show that $f(x) + f\left(\dfrac{1}{x}\right) = 0$

4. If $f(x) = \dfrac{x+1}{x-1}$, $x \neq 1$, then prove that $f(f(x)) = x$

5. For $f(x) = \dfrac{x-1}{3x+1}$, $x > 1$, write the expressions of $f\left(\dfrac{1}{x}\right)$ and $\dfrac{1}{f(x)}$

6. If $f(x) = e^x$ and $g(x) = \log_e x$, then find

   (i) $(f+g)(1)$ (ii) $(fg)(1)$

   (iii) $(3f)(1)$ (iv) $(5g)(1)$

7. Draw the graph of the following functions:

   (i) $f(x) = 16 - x^2$

   (ii) $f(x) = |x - 2|$

   (iii) $f(x) = x|x|$

   (iv) $f(x) = e^{2x}$

   (v) $f(x) = e^{-2x}$

   (vi) $f(x) = \dfrac{|x|}{x}$

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